PHASE 3: PROJECT DESIGN PHASE

EV Market Performance & Insights

Date	20 – October - 2026
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Project Name	Electric vehicle Market Performance & Insights

Problem Solution Fit • Proposed Solution • Solution Architecture

Purpose: This document describes the fit between the user problem and the proposed Tableau solution, outlines the solution workflow, and presents the high-level architecture. It follows the three Project Design headings listed in the NASSCOM phase checklist. If your trainer has a separate blank template, match any additional mandatory fields before submission.

1. Problem Solution Fit

1.1 Problem Context

The available electric vehicle dataset contains multiple dimensions and measures, including EV category, model year, manufacturer, battery technology, annual sales units, driving range, customer ratings, and manufacturing country. Comparing these fields individually can make it harder for a user to build a consolidated view of the dataset.

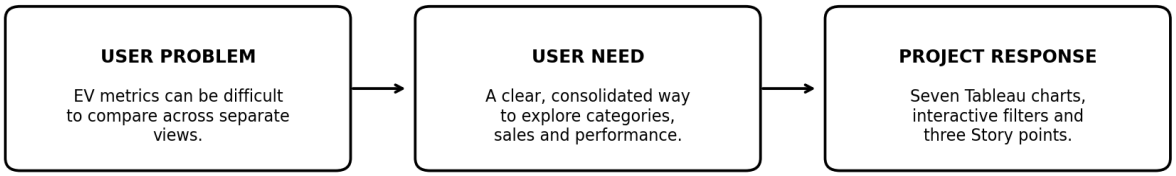
1.2 User Need

An analyst or business user needs a clear and interactive way to compare EV-related categories and measures, focus on selected segments, and communicate observations without manually reviewing separate tables or charts.

1.3 Proposed Fit

The Tableau-based solution addresses this need by organizing the available EV data into seven visualizations in a dashboard, providing available filters, and arranging related views into three Story points. This supports descriptive exploration of the dataset.

PROBLEM-SOLUTION FIT



FIT CRITERIA

Relevant to EV questions • Feasible with the available dataset • Easy to explore • Findings can be verified

1.4 Fit Assessment

Criterion	How the project addresses it	Assessment
Relevance	Charts address category, sales, manufacturer, battery, range, ratings, and country questions.	Aligned with the stated project scope.
Feasibility	Uses the available dataset and Tableau workbook.	Feasible for a Tableau-focused project.
Usability	Dashboard combines views and exposes filters for exploration.	Requires readability and filter testing.
Evidence	Observations can be checked against chart values and the dataset.	Use only verified values in final findings.
Scope	Descriptive visual analysis, not forecasting or live monitoring.	Keep claims within available data.

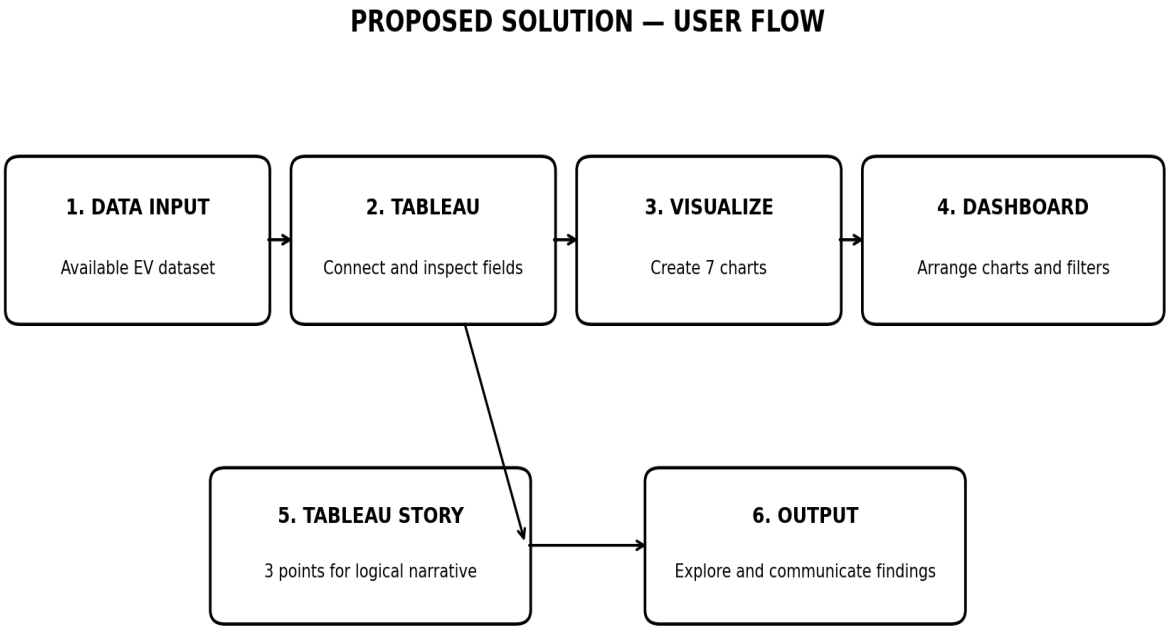
1.5 Expected Benefits

- A consolidated overview of the selected EV market dimensions.
- Faster visual comparison across categories and manufacturers.
- Interactive exploration using the filters available in the dashboard.
- A structured narrative through three Tableau Story points.
- Clearer communication of data-supported observations in the report and demo.

2. Proposed Solution

2.1 Solution Overview

The proposed solution is an interactive Tableau dashboard titled “EV Market Performance & Insights”, supported by a Tableau Story. It presents seven charts covering EV category performance and distribution, annual sales trends by model year, manufacturer comparison, battery technology distribution, average driving range by category, customer ratings by category, and manufacturing-country distribution.



2.2 Main Components

Component	Purpose	Project-specific details
Dataset	Provides the data for analysis.	Use the EV dataset connected to the final Tableau workbook; verify whether train/test files or another file is the active source.
Seven visualizations	Answer descriptive business questions.	Category, annual sales trend, manufacturer, battery technology, range, customer ratings, country distribution.
Dashboard filters	Allow focused exploration.	Category, manufacturer, and model-year filters are visible in the dashboard screenshot; test their behavior.
Tableau Story	Presents views in a logical sequence.	EV Market Overview; Vehicle & Battery Insights; Manufacturer & Country Insights.
Documentation	Explains process and findings.	Business Question & Visualisation Report, Dashboard Design, Story Design, performance testing, and final report.

2.3 User Workflow

- Open the Tableau workbook and view the EV market overview.
- Read the chart titles and compare the measures shown.
- Use available filters to focus on selected categories, manufacturers, or model years.
- Navigate through the three Story points to explore related insights.
- Record only those findings that can be verified from the visible charts and source data.

2.4 Scope and Limitations

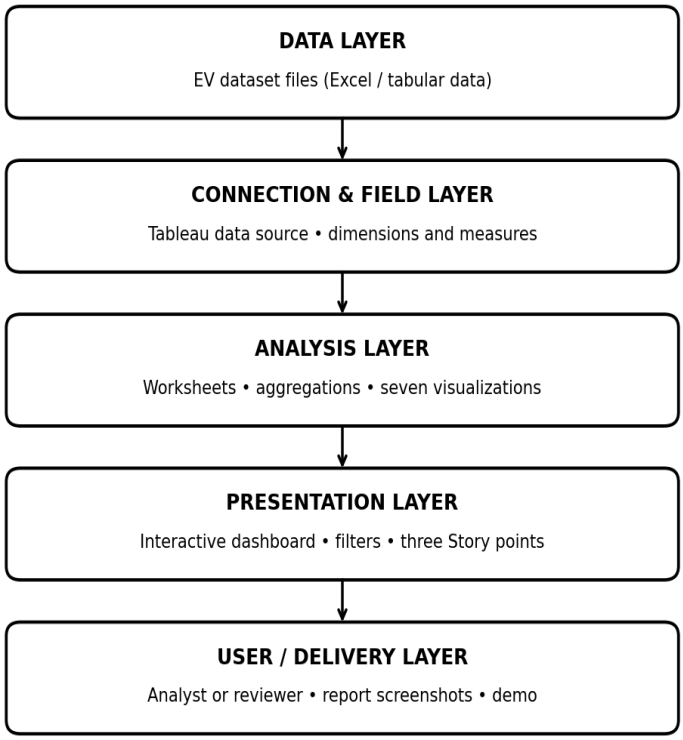
- Included: descriptive visualization, interactive dashboard exploration, and Tableau Story presentation.
- Not included unless separately implemented: predictive modelling, live market data feeds, automated alerts, or external API integration.
- Results are limited to the fields, records, definitions, and time period in the supplied dataset.

3. Solution Architecture

3.1 High-Level Architecture

The solution follows a simple data-to-visualization workflow. The dataset is connected to Tableau, fields are used in worksheets to build charts, the charts are arranged into a dashboard with filters, and related views are organized into a Story for presentation.

SOLUTION ARCHITECTURE — TABLEAU PROJECT



3.2 Architecture Layers

Layer	Description	Output
Data Layer	EV dataset file(s) available to the project.	Tabular source data.
Connection & Field Layer	Tableau data source exposes dimensions and measures.	Fields ready for chart construction.
Analysis Layer	Worksheets use selected fields and aggregations to form seven visualizations.	Individual charts.
Presentation Layer	Dashboard combines the charts and available filters; Story organizes views into three points.	Interactive dashboard and Story.
User / Delivery Layer	Users review the dashboard; screenshots and findings are used in reports and demo.	Project evidence and communication.

3.3 Data and Interaction Flow

- The EV dataset is connected as a Tableau data source.
- Relevant dimensions and measures are selected for each worksheet.
- Tableau visualizations summarize the data using the configured measures and aggregations.
- The dashboard combines the worksheets and provides available filters.
- The Tableau Story arranges the analysis into three points.
- The user explores the charts and communicates data-supported observations.

3.4 Architecture Assumptions

- This is a high-level architecture of the current Tableau-focused project.

- No database server, cloud pipeline, API, or live-refresh process is claimed unless separately configured.
- The final workbook should be opened and checked to confirm its connected data source and available filters.